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JAVA LOOP PRACTICE WORKBOOK

Phase 1 Extension: Logic Boot Camp for 2027 Goals

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CATEGORY 1: DIGIT MANIPULATION (Building on Reverse & Count Mechanics)

PROBLEM 26: Sum of Digits

- Goal: Take an integer input from the user and calculate the sum of all its individual digits.
- Example Input: 1234
- Example Output: Total Sum of Digits: 10 (Since $1 + 2 + 3 + 4 = 10$)
- Target Package: Day3_Standard_Iteration_Loops
- Interview Note: Tests basic mastery of using % 10 to extract and / 10 to clear.

PROBLEM 27: First and Last Digit Sum

- Goal: Given any integer, find the sum of only its first (most significant) digit and its last (least significant) digit.
- Example Input: 5674
- Example Output: First & Last Sum: 9 (Since $5 + 4 = 9$)
- Target Package: Day3_Standard_Iteration_Loops
- Logic Hint: The last digit is easily found via 'num % 10'. To find the first digit, use a while loop that keeps dividing by 10 until 'num' is less than 10.

CATEGORY 2: MATHEMATICAL SERIES & CONDITIONALS

PROBLEM 28: Perfect Number Check

- Goal: A perfect number is a positive integer that is equal to the sum of its positive divisors, excluding the number itself. Write a program to check if an input is perfect.
- Example Input: 6
- Example Output: 6 is a Perfect Number.
(Divisors of 6 are 1, 2, 3 -> $1 + 2 + 3 = 6$)
- Target Package: Day4_Complex_Loop_Logic

PROBLEM 29: Print Prime Numbers in a Range

- Goal: Ask the user for a starting integer and an ending integer, then output all prime numbers situated between that range (inclusive).
- Example Input: Start = 10, End = 20
- Example Output: Prime numbers: 11, 13, 17, 19
- Target Package: Day4_Complex_Loop_Logic
- Logic Hint: This requires nested loops. The outer loop moves from number to number, and the inner loop checks if the current number is prime.

CATEGORY 3: ADVANCED NESTED LOOPS & GRID MECHANICS

PROBLEM 30: Left-Aligned Inverted Triangle (With Spaces)

- Goal: Print an inverted triangle made of stars, but aligned to the right-hand margin of the terminal screen window.
- Target Package: Day5_Nested_Loops_Patterns_Advanced_Math
- Logic Hint: You need three loops. An outer loop for rows, a sub-inner loop to print incremental spaces, and a second sub-inner loop to print decremental stars.
- Expected Output Shape:

```
*****
****
***
**
*
```

PROBLEM 31: Number Pyramid (The Classic Pattern)

- Goal: Print a visually balanced, centered pyramid compiled of consecutive row-based numbers.
- Target Package: Day5_Nested_Loops_Patterns_Advanced_Math
- Expected Output Shape:

```
1
1 2
1 2 3
1 2 3 4
```

CATEGORY 4: STATE TRACKING & CONDITIONAL CONTROL STREAMS

PROBLEM 32: Find the Largest and Smallest Input (Sentinel Loop)

- Goal: Continually request the user to type integers. Keep looping infinitely until the user enters '-1'. Once '-1' is typed, stop the loop and print out the highest and lowest numbers entered during the entire session.
- Example Run: Inputs: 5, 12, 3, 44, 2, -1
- Example Output: Largest: 44, Smallest: 2
- Target Package: Day3_Standard_Iteration_Loops
- Logic Hint: Use a 'while(true)' loop with an internal 'if (input == -1) break;' mechanism.

PROBLEM 33: LCM (Lowest Common Multiple)

- Goal: Accept two integers from the user and calculate their Lowest Common Multiple.
- Example Input: n1 = 12, n2 = 18
- Example Output: LCM is 36
- Target Package: Day4_Complex_Loop_Logic
- Formula: $LCM = (n1 * n2) / GCD$
- Logic Hint: Reuse your Problem 20 GCD subtraction loop to find the answer efficiently!

CATEGORY 5: BASE CONVERSIONS (The MNC Developer Hurdles)

PROBLEM 34: Binary to Decimal Conversion

- Goal: Take a binary sequence integer (composed of only 0s and 1s) and convert it into its base-10 base integer value using a custom loop.
- Example Input: 1011
- Example Output: Decimal Value: 11
- Target Package: Day4_Complex_Loop_Logic
- Math Tracking: $(1 * 2^3) + (0 * 2^2) + (1 * 2^1) + (1 * 2^0) = 8 + 0 + 2 + 1 = 11.$

PROBLEM 35: Decimal to Binary Conversion

- Goal: Complete the inverse mechanism—accept a decimal integer input and calculate its equivalent binary representation string using loops.
- Example Input: 11
- Example Output: Binary Value: 1011
- Target Package: Day4_Complex_Loop_Logic
- Logic Hint: Keep dividing the number by 2 using a loop while tracking and appending the remainders (% 2).

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